Yangen Li

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Education

Ph.D. in Finance, University of Iowa	2027 (expected)
M.Sc. in Quantitative Financial Economics, Oklahoma State University	2021
M.Sc. in Chemical Engineering, China University of Petroleum	2019
B.A. in Chemical Engineering, Beijing University of Chemical Technology	2016

Research Interests

Financial misconduct and disclosure; executive contracting; machine-learning and LLM-based measurement; empirical asset pricing

Job Market Paper

Future Proves Past: Detecting Hidden Fraud under Imperfect Enforcement Labels, with C. Wei Li, Yuxuan Li, and Anand M. Vijh.

- *Detects hidden fraud that SEC enforcement never surfaces, validated entirely outside the enforcement label*
- *Custom residual CNN on multi-year accounting panels with a certainty-weighted focal loss over soft supervisory labels*
- 2026 University of Iowa Finance Brown Bag Seminar

Working Papers

Listening to History Rhymes: Pattern Recurrence and Return Predictability in Stock Markets, with C. Wei Li.

- *Pattern recurrence in return trajectories is persistent, predicts returns, and survives transplanting to twelve international markets*
- *Hundreds of billions of pairwise trajectory distances computed on GPU to map the full cross-section*
- 2026 Financial Management Association (FMA) Annual Meeting (scheduled); 2026 Silicon Prairie Finance Conference; 2024 University of Iowa Finance Brown Bag Seminar

CEO Individualism and the Distribution of Performance Risk, solo-authored.

- *CEOs are placed on an individualism–collectivism spectrum using their conference-call language, with Q&A exchanges distinguished from prepared statements*
- *Individualistic CEOs pass performance shocks through to employment at roughly twice the average CEO's rate—only in downturns*

The Protestant Ethic and the Principal-Agent Relation: Evidence from CEO Employment Agreements, with C. Wei Li, Yuxuan Li, and Yiming Qian.

- *Firms in more Protestant counties write fewer and thinner CEO contracts; the effect is denominational, not devotional, as the same result does not appear in more Catholic counties*
- 2025 Eastern Finance Association (EasternFA) Annual Meeting

When "Milliseconds" Matter: The Real Effects of High-Frequency Trading, with Wesley Deng, Bo Hu, and Amy Kwan.

- *High-frequency trading lowers corporate investment and equity issuance and raises the implied cost of equity, most for firms that rely on external equity*
- *Identified from the 2012 launch of the ASX-ITCH low-latency data feed with broker-level HFT identifiers; corroborated by the staggered global rollout of exchange co-location*
- 2025 FMA Annual Meeting New Idea Session; Chinese University of Hong Kong; George Mason University

Methods Paper

Building a Contract Dataset with a Fine-Tuned Language Model, and What a Commercial Database Measures, solo-authored.

- *Fine-tunes an open-weight language model to read the full EDGAR exhibit record, proposes a protocol for validating datasets built by language models, and shows that a leading commercial database's employment-agreement flag measures the proxy-statement description rather than the filed contract*
- Code: github.com/yangenli/llm-finetuning

Teaching Experience

University of Iowa, Tippie College of Business
Stand-alone Instructor

- Introductory Financial Management, Spring 2024
- Investment Management, Summer 2025 & Summer 2026

Teaching Assistant

- Introductory Financial Management (Discussion Session ×3), Spring 2023

Grants & Awards

Graduate College Travel Award, University of Iowa	2024–Present
Ponder Fellowship, University of Iowa	Summer 2025
Graduate College Summer Fellowship ($5,500), University of Iowa	2024–2025
Graduate College Post-Comprehensive Research Fellowship ($11,000)	Feb.–June 2025
Spears School of Business Full Graduate Scholarship, Oklahoma State University	2019–2021

Dissertation Committee Members

C. Wei Li (Chair)
Associate Professor of Finance
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Ashish Tiwari
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Anand M. Vijh
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Additional Information

Computing

- Fine-tune and distill open-weight LLMs (QLoRA) on a self-built three-GPU workstation (88 GB VRAM); all training and inference run locally—used to classify over one million SEC filings and to compute billion-scale trajectory-distance matrices

Programming & Software

- Python, SAS, Stata, R
- CLI coding agents

Languages

- English (fluent); Mandarin (native)

Abstracts

Future Proves Past: Detecting Hidden Fraud under Imperfect Enforcement Labels, with C. Wei Li, Yuxuan Li, and Anand M. Vijh.

Abstract: SEC enforcement labels are selective and incomplete proxies for the latent fraud they are meant to measure. This mismatch creates two problems: a model trained on them may learn the enforcement process rather than the fraud process, and accuracy measured against them cannot reveal whether a model detects the fraud enforcement misses. We address both with a *future-proves-past* design. Because accounting distortions eventually reverse, future accounting outcomes modestly refine the training signal into soft labels; we train a convolutional neural network on multi-year accounting panels with these labels and validate it entirely outside the enforcement set. The model gives up little SEC-labeled detection performance. Out of sample over 2000–2019, flagged firms complete seasoned equity offerings at nearly three times the base rate and earn large negative post-issuance abnormal returns, a long-short portfolio earns significant factor-adjusted alpha, and externally screened populations (post-scrutiny Arthur Andersen clients and bank-monitored borrowers) are systematically avoided. An identically specified network trained on the enforcement label, and benchmarks designed to predict it, reproduce little of this profile beyond a size-controlled tilt away from bank borrowers: the training target drives the difference. The results show that selective enforcement data can be repurposed, at modest cost, to reach fraud that enforcement itself never surfaces, and they caution that detection models should be judged by validation outside the very label on which they were trained.

Listening to History Rhymes: Pattern Recurrence and Return Predictability in Stock Markets, with C. Wei Li.

Abstract: Studies of price patterns have always had to say in advance which patterns to look for. We let the data say instead. We cut the daily return history of U.S. equities from 1970 to 2025 into roughly 1.32 million overlapping 60-day segments, standardize each one, and compare it against every other. A segment recurs when the real data are more crowded around it than a matched random-walk benchmark would imply, and the ratio of the two—the *Pattern Distance Ratio* (PDR)—turns recurrence into a continuous property of a return path that can be measured and tested. Recurrence is genuine and persistent: a segment's PDR in one year predicts its PDR in the next at a correlation of 0.308, still 0.270 three years out, and positive in every year of the sample, against 0.011 for a random-walk placebo. Recurrence also predicts returns. Segments that recur strongly carry more reliable directional forecasts, and a kernel ridge strategy built in trajectory space earns gross long–short alphas of roughly 6% to 9.5% per year against the CAPM and the Fama–French three- to six-factor models. The alphas hold in the most recent subperiods, and technical rules and standard predictors do not account for them. Deployed on twelve international markets with every design choice frozen at its U.S. value, the strategy earns a positive raw equal-weighted spread in all twelve, and a global country-average portfolio earns a significant alpha.

CEO Individualism and the Distribution of Performance Risk, solo-authored.

Abstract: I measure CEO individualism using only the CEO's answers to analyst questions on earnings calls—answers delivered without a prepared script, and so less filtered than the remarks that open the same call—projecting each sentence onto a transformer-based individualism axis built from 100 validated anchor sentences and dropping answers to narrow numerical and procedural questions. In an ExecuComp firm-year panel, 2010–2023, a one-standard-deviation increase in CEO individualism raises the pass-through of a one-unit ΔROA shock to employment by 7.3 percentage points. The effect is concentrated in downturns: individualistic CEOs cut workforces aggressively, while the good-times differential is positive but imprecisely estimated. I call this a *scarcity-revealed weighting* channel: the relative weight a CEO places on shareholder returns versus employee welfare is observable from outcomes only when scarcity forces a binding tradeoff. The asymmetry favors this channel over a symmetric efficiency story, though equality of the two interactions is rejected only at the 10% level. A separation-composition margin agrees: in downturns, more of the separations at individualistic CEOs' firms are of the abrupt kind least likely to be voluntary, though the same tilt appears more weakly in upturns, marginally significant rather than a clean null.

The Protestant Ethic and the Principal-Agent Relation: Evidence from CEO Employment Agreements, with C. Wei Li, Yuxuan Li, and Yiming Qian.

Abstract: We ask whether the religious culture surrounding a firm's headquarters shapes the most basic formal instrument of the principal–agent relation: the written employment agreement between a firm and its chief executive. Using large language models to read exhibit filings from the SEC's EDGAR system, we assemble 6,373 explicit CEO employment contracts, of which 4,660 are fixed-term employment agreements (FEAs), covering 3,468 CEOs at 2,068 Execucomp firms between January 2005 and December 2022—to our knowledge the most comprehensive sample of CEO employment contracts assembled to date. Firms headquartered in counties with higher Protestant adherence are markedly less likely to use FEAs, and the coefficient on Protestant adherence is negative and significant at the one percent level in every specification. Two alternative measures of contractual explicitness—an index built from ten key compensation provisions and the log length of the contract text—deliver the same negative relation, so agreements written in more Protestant counties are not merely rarer but thinner. The effect is denominational rather than devotional: total religious adherence carries no explanatory power, whereas Catholic adherence enters positively and significantly, so the two traditions are associated with contractual formality of opposite sign. Extending the analysis to forced CEO turnover yields directionally consistent but statistically insignificant evidence of greater board forbearance.

When "Milliseconds" Matter: The Real Effects of High-Frequency Trading, with Wesley Deng, Bo Hu, and Amy Kwan.

Abstract: This paper examines how high-frequency trading affects corporate investment and financing. Using a unique Australian setting with broker-level HFT identifiers, we exploit the introduction of ASX-ITCH, a high-speed order-level market data feed that increased the speed of trading information, as plausibly exogenous variation in HFT presence. As HFT activity rises, firms experience a sizable decline in investment and equity issuance. These effects are concentrated among firms that rely more heavily on external equity financing. We find corroborating evidence in the staggered global rollout of exchange co-location services. Overall, the findings show that financial market innovations can alter firms' financing capacity and investment decisions, providing a channel through which financial market structure may influence aggregate economic growth.

Building a Contract Dataset with a Fine-Tuned Language Model, and What a Commercial Database Measures, solo-authored.

Abstract: I fine-tune an open-weight 14-billion-parameter language model on 20,000 machine-labeled SEC exhibits and use it to read the EDGAR exhibit record from 2005 through June 2026. That is 1.38 million documents, classified on three consumer GPUs. What fine-tuning buys is precision. On held-out documents the tuned model reaches an F_1 of 0.978, against 0.919 for the best prompt I found, and it makes 27 false positives where prompting makes 140. But a classifier that reads documents well does not by itself give a correct dataset. I propose a way to check one: draw stratified samples, judge them against an independent source, read every case twice, and plant control cases whose answer is known. On 410 cases, almost none of the remaining error came from misreading documents. It came from the code that decides which person and which year a document belongs to. The same check, applied to the leading commercial database, shows that its employment-agreement flag and mine do not measure the same thing. Among CEOs it never credits with an agreement, 21.8 percent have a formal employment agreement on file. And when the document is an offer letter, the vendor calls it an agreement only 57.7 percent of the time. Firms must file an executed contract, but they choose how to describe it. A measure built from filings sees the document. A measure built from proxy statements sees the description.